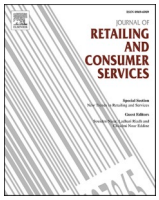
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I am not like them: A terror management theory perspective on the consumer separation tendency of pet profile images

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ABSTRACT

Despite increasing trends of influencers using pet elements to pursue business benefits, little attention has been paid to understanding their harmful effects. According to the Terror Management Theory, this research aimed to fill this gap by exploring how mortality salience reduced the effectiveness of influencers with pet profile images. On the basis of four experiments, we found that mortality salience hurt consumer's attitude toward them because increased anxiety about physical death led consumers to perceive less similarity and separate from pet images. Furthermore, the above negative effect on consumer's attitude was reversed by their posts disclosing worldviews (a sense of meaning transcending death), which provided consumers with an effective defense tool against fear of mortality.

1. Introduction

In recent years, social media has been filled with information that can be considered cues for mortality salience, such as celebrity death (Death of Queen Elizabeth II, 2022), earthquake (Turkey-Syria earthquake, 2023), and so on (Lyu et al., 2023). Mortality salience refers to reminding individuals of their mortality and the understanding that death is inevitable (Rahimah et al., 2020). As mortality-salient news becomes increasingly prevalent in our daily lives, the way information is presented on social media platforms can subtly influence consumers' attitudes (Barnes, 2021; Kwon and Park, 2022).

In the social media context, consumers rely on influencers, who have become one of the most powerful driving forces for consumer attitudes (Venciute et al., 2023; Wang and Liu, 2023). Particularly, the profile image of influencers refers to the accompanying image that appears with the influencer's name and the first picture that potential audiences see before they watch their posts content. Previous research claimed that profile pictures reflected influencers' self-image and self-expression (Hu et al., 2022), which is crucial for their impression management (Jang, 2022; Kim et al., 2020) and eventually contributes to audiences' likability and trustworthiness (Hu et al., 2022). Among the different kinds of profile images, there has been a growing trend of using pet images to attract audiences and boost businesses on social media over recent years. For example, as pet marketing grows, influencers using pet profile images correspondingly offer brands many opportunities, which are anticipated to reach over $200 billion globally by 2025 (Myers et al., 2024). Previous research demonstrated that pet images offer an effective measure of promoting products and services since these "fur babies" provide the audience with emotional belonging like their family members (Myers et al., 2024). However, do using pet profile images always benefit businesses? There is scant research that examines whether using pet profiles might bring negative impacts on endorsing effects. As mentioned in the beginning, at the same time, social media has been filled with information that can be considered cues for mortality salience. It inspires us to consider whether consumers tend to separate from influencers with pet profile images when faced with mortality salience, as prior research suggested that animals symbolize the shared biological fate and vulnerability that humans wish to deny (Goldenberg et al., 2001; Lifshin et al., 2023). Unfortunately, scant research has focused on the effect of mortality salience on consumer attitudes towards influencers using pet profile images. Given the identified gap, it is critical to address the following research questions.

RQ1: How and why does mortality salience affect consumer attitudes toward influencers using pet profile images?
RQ2: Under what conditions does the effect of mortality salience on consumer attitudes intensify or diminish?

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RQ3: What strategies can reverse the negative effect of mortality salience on consumer attitudes toward influencers using pet profile images?

Toward this, the current work aimed to investigate when and why negative effects of pet profile images appeared and further discussed strategies to reverse negative effects based on the terror management theory (TMT) (Greenberg et al., 1992). Specifically, we found that mortality salience would negatively affect consumer attitudes towards influencers using pet profile images. This is due to fear of death during the mortality salient period, which motivates consumers to deny their animality (the same biological conditions as other animals), eventually impairing their perceived similarity to influencers using pet profile images and then contributing to negative attitudes about products endorsed by them. This consumer reaction is more intense when consumers regard themselves as more analogous to animals (high perceived creatureliness). Further, we explored worldview disclosure derived from TMT reversing this negative effect. Since worldview links the self toward something greater and more permanent than the physical self (Dunn et al., 2020; Lyu et al., 2023), which assuages mortality anxiety.

Based on these findings, we made several contributions. First, we proposed the negative effect of using pet profile images when considering a new antecedent of mortality salience. Recent research regarded pet images as an effective means to promote consumer engagement because of people's metaphoric relationships and perceived ownership with pets (D'Souza et al., 2023). However, they ignored cues for salient mortality, which is ubiquitous in current social media context. Based on TMT, our research provided evidence that such cues may lead pet images to be less effective. The reason why TMT is relevant to influencer marketing is that TMT-related literature indicated that reminding of mortality salience increased people's need to distance from biological identity as one of psychological defenses to death anxiety (Lifshin et al., 2023; Liu et al., 2022). Furthermore, this tendency, separate from biological identity, may lead consumers to perceive less similarity with pet images, thereby impairing their attitudes towards influencers using pet images.

Second, this research also advances knowledge of applying the TMT in influencer marketing by examining a novel mechanism and the boundary condition. In recent decades, mortality salience has widely been examined in the field of marketing, such as overconsumption (Mandel and Smeesters, 2008); donation behavior (Bock et al., 2018); green behavior (Rahimah et al., 2020); AI service (Liu et al., 2022); heritage tourism (Lyu et al., 2023); and so on. However, to our best knowledge, scant prior research considered mortality salience in pet influencer marketing. Particularly, the current work addresses this gap while also uncovering a novel mechanism of how consumers' perceived similarity with animals plays a role between mortality salience and consumers attitude towards influencers using pet images. Furthermore, this research proposes the boundary condition of a cultural worldview derived from TMT. In so doing, it also adds to the recent discussion of how the worldview contributes to reversing the negative effect of mortality salience in consumer attitude (Lyu et al., 2023; Rahimah et al., 2020).

Finally, this research possesses applied implications. Derived from the conception of cultural worldview in TMT (Greenberg et al., 1992), we proposed feasible strategies—worldview disclosure (e.g., disclosing donation activities), encouraging people to seek the meaning in life, which provides the effective tool to defend against anxiety of physical death, thereby reversing the original negative attitude caused by mortality salience towards influencers using pet profile images. This offers novel insights into pet influencer marketing practices.

2. Theoretical background

2.1. Influencer marketing

Influencers are individuals who have established a network of followers on social media and are considered to have significant social influence over their network of followers (Leung et al., 2022). Specifically, social media profile image choice becomes one of the important factors since profile images were found to reflect self-presentation and impression management of influencers (Abbasi et al., 2023). Self-presentation involves sharing one's information to create impressions for others, which reveals certain pieces of information and characteristics of influencers (Schlosser, 2020). Furthermore, a successful self-presentation of influencers can draw benefits from their viewers. For example, cute profiles are treated as a positive and warm trait cue, contributing to audiences' likability and trustworthiness towards influencers (Hu et al., 2022). However, there is scant research exploring how using pets as profile images affects the effectiveness of influencers.

In fact, previous research has demonstrated why pet influencers could be an effective way to attract consumers. Pet, as part of ourselves, plays important roles in consumers' daily lives (Jia et al., 2022). They provide consumers with a metaphoric relationship and emotional belonging (Jeong et al., 2024; Zhang et al., 2023). Thus, it is supposed to infer that using pets as profile images will bring a positive effect on the effectiveness of influencers. However, recent research demonstrated that considering the transgression context, ads with an animal-like image (vs. a human image) provoke a more intense moral indignation reaction because of the perception of dehumanization (Huang and Hsieh, 2023). This implied the pet image may bring more negative emotion to consumers under certain situations. The current research investigates another social context (i.e., mortality salience) that would weaken the effectiveness of using pet profile images, which has been scantly explored in previous research.

2.2. Mortality salience and terror management theory

Terror management theory (TMT) posits that the consciousness of mortality is troublesome as it clashes with the fundamental instinct to survive (Greenberg et al., 1992). Therefore, it creates a potential for existential terror. TMT asserts that, when faced with the terror of mortality salience, humans have developed a defensive psychological system that allows concerns about death to leave consciousness. Over the past few decades, TMT has been utilized in the domain of consumer behavior alterations triggered by mortality salience. For instance, prior studies have examined that when mortality salience is activated, consumers are inclined to engage in excessive consumption and increased donations as a form of defense to evade self-awareness and to strive for social desirability (Cai and Wyer, 2015; Mandel and Smeesters, 2008). Recent research found consumers tend to be hesitant to embrace humanoid (as opposed to non-humanoid) service robots when the awareness of mortality is heightened because a human-like appearance threatened consumers' human identity and sparked a defensive mechanism towards mortality salience (Liu et al., 2022). Inspired by this perspective, this research further explores how consumers respond to broader biological images, not limited to human-like images, when facing mortality salience.

Furthermore, in line with TMT, individuals possess a crucial coping mechanism when confronted with death-related anxieties—their worldview. A worldview encompasses shared norms and convictions that contribute to the establishment of a sense of order, significance, and immortality. (Greenberg et al., 1992). Worldviews delineate pathways to transcendence in the face of mortality awareness, enabling individuals to attain symbolic immortality by aligning themselves with entities that are more extensive and enduring than their physical existence (Burling, 1993; Dunn et al., 2020). Recent marketing research investigated the critical role of worldview on consumer behaviors when facing mortality salience. For example, recent research investigated that to reduce the existential insecurity caused by mortality salience, individuals may enhance their sense of value within their own cultural worldview through the acquisition of high-status products or the pursuit of materialism, which in turn helps to mitigate the increased

accessibility of thoughts related to mortality (Rahimah et al., 2020). Besides, another study discovered that with the rise in individuals' general anxiety about mortality, their inclination to seek meaning in life also grows. This, in turn, promotes their preference for heritage travel and their intention to protect cultural heritage. (Lyu et al., 2023). Following this, this research clarified how influencers using pet profiles could reverse the negative effect triggered by mortality salience by providing consumers with a cultural worldview disclosure.

According to Terror Management Theory, mortality salience heightens individuals' awareness of their physical limitations and the inevitability of death, triggering fear and anxiety related to their biological vulnerability (Cox et al., 2007). To counter these death-related thoughts, people develop intrinsic defense mechanisms (Cheng et al., 2024). Prior research has shown that humans tend to distance themselves from animals when faced with mortality salience, as animals symbolize the shared biological fate and vulnerability that humans wish to deny (Goldenberg et al., 2001; Lifshin et al., 2023). In essence, the similarities between humans and animals intensify concerns about bodily existence, leading individuals to reject such similarities and distance themselves from animal representations. In the context of pet influencer marketing, this implies that mortality salience diminishes consumers' perceived similarity with pet images, as such images may remind them of their biological connection to animals.

Further, according to similarity-attraction theory, perceived similarity can lead to interaction, attraction, and positive attitudes (Bu et al., 2022; Xu et al., 2021). People are more inclined to be attracted to others who share similar characteristics, as these similarities reinforce their worldviews and satisfy their psychological need for coherence and self-validation (Byrne, 1961). In other words, people who are similar to us tend to reinforce our sense of self, which is associated with positive emotions, and this ultimately results in feelings of attraction (Cheng, 2025). Conversely, when consumers perceive less similarity with influencers, especially those who may challenge or fail to align with their worldview— they are more likely to form negative attitudes toward such influencers, in so doing, reinforcing the logic and consistency of our world. Therefore, mortality salience has a negative effect on consumer attitudes toward influencers using pet profile images, which is mediated by perceived similarity. Thus, we proposed that.

H1(a). Mortality salience has a negative effect on consumer attitudes toward influencers using pet profile images.

H1(b). Perceived similarity mediates the relationship between mortality salience and consumer attitudes toward influencers using pet profile images.

Furthermore, we explored the moderating role of perceived creatureliness on the above relationships between consumers' attitudes and mortality salience. When consumers felt higher creatureliness concerns while facing mortality salience, they believed that the boundaries between them and animals were blurred. That is, humans and animals had similar creaturely limitations, i.e., inevitable death. Further, these consumers tend to develop emotional rejection toward vulnerability to death (Goldenberg et al., 2000; Lifshin et al., 2023). Therefore, consumers feeling higher creatureliness generate more separation and more negative attitudes toward influencers using pet profile images (Zaleskiewicz et al., 2015). In contrast, consumers who perceive lower creatureliness believe they are different from other animals, thereby being less likely to think about bodily limitations and death anxiety even in certain conditions of mortality salience (Cox et al., 2007), thereby buffering the negative attitudes towards influencers using pet profile images. Thus, we proposed H2.

H2. Perceived creatureliness moderates the negative relationship between mortality salience and consumer attitude towards influencers using pet profile images.

H2 (a). When perceived creatureliness is high, mortality salience will bring a larger negative impact on consumer attitude.

H2 (b). When perceived creatureliness is low, mortality salience has no significant impact on consumer attitude.

Finally, we explored the effect of one defense mechanism proposed by the TMT-worldviews on the connection between mortality salience and endorsement outcomes of influencers using pet profile images. According to TMT, supporting worldviews in response to mortality salience enables people to achieve symbolic immortality (Zaleskiewicz et al., 2015), which overcomes people's fear of creaturely limitations and physical death. Following this, when influencers using pet profile images provide worldview disclosures, such as disclosing donation activities, it could trigger people to think in a broader way and allow in meaning for life, enabling the individual to maintain a symbolic existence beyond their physical being. (Lee et al., 2023). Such thoughts attenuate their anxiety of physical death, thereby buffering the original negative effect of mortality salience and bringing a positive impact on consumer attitude toward influencers using pet profile images. Thus, we proposed H3. The theoretical model is presented in Fig. 1.

H3. Worldview disclosure moderates the effect of mortality salience on consumer attitudes toward influencers using pet profile images. Specifically:

H3 (a). When a worldview disclosure is present, the negative effect of mortality salience on consumer attitudes will be mitigated or reversed.

H3 (b). When a worldview disclosure is absent, the negative effect of mortality salience on consumer attitudes will persist.

3. Overview of studies

The hypotheses are examined through four experiment-based studies, each having its own specific objectives detailed below.

- Study 1a aimed to test whether under the mortality salience condition (vs. control condition), influencers using pet profile images will lead to decreased consumer attitude through the mediation of perceived similarity compared to the counterparts—H1(a) and H1 (b). Specifically, we measured consumer attitude using two dimensions: product attitude and purchase intention, in line with prior research (Alcántara-Pilar et al., 2024; Liu & Liu, 2019, 2020).
- Study1b tested that mortality salience brings negative consumers' implicit associations with influencers using pet profile images using the IAT method—H1(a).
- Study 2 manipulated the theoretical moderator (perceived creatureliness) and enhanced the evidence that it is when there are higher perceived human-animal similarities that mortality salience causes denying animality and thereby negative consumer attitudes towards influencers using pet profile images (H2).
- Study 3 assessed the moderating role of worldview disclosures reversing the effect of mortality salience on influencing outcomes (H3).

Tables 1 and 2 respectively show the demographic information and scales (perceived similarity, product attitudes and purchase intention) reliability and validity of each experiment. The Cronbach's alphas of all scales are higher than the critical value of 0.70, and the values of CR (Composite Reliability) are greater than the cut-off of 0.70 (Hair et al., 1986; Zhang et al., 2024).The values of AVE (Average Variance Extracted) are greater than the critical threshold of 0.50 (Hair et al., 1986; Zhang et al., 2024), except for purchasing intentions in Study1a (0.492), which is almost 0.50. Therefore, it indicates that reliability and validity are verified. In addition, we have done four replication studies with representative and balanced samples for Study 1A, Study 1B, Study 2, and Study 3, respectively, to ensure the robustness and generalizability of our findings (Online Appendix).

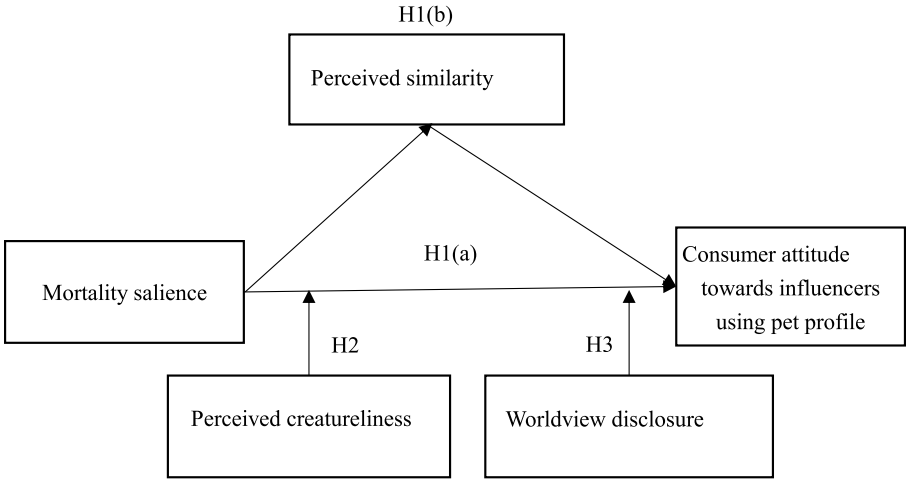


Fig. 1. Theoretical model.

Table 1
Participants' demographic information.

	Study1a	Study1b	Study2	Study3
Gender				
Male	93	79	135	134
Female	115	67	146	197
Age				
18 to 25	36	7	51	55
26 to 30	98	71	129	136
31 to 40	59	56	88	127
41 and above	15	12	13	13
Education				
High school and less	9	6	8	9
Associate College	18	15	25	38
Bachelor	169	110	223	261
Master and above	12	15	25	23
Income (per month)				
<2k	11		17	20
[2k,5k)	48		39	43
[5k,10k)	86		139	158
[10k,20k)	54		73	84
≥20k	9		13	26
Total	208	146	281	331

3.1. Study 1a mortality salience and influencers using pet (vs. Non-pet) profile images

3.1.1. Method

Participants. A total of 240 participants were recruited from Wenjuanxing. Those who did not write according to the prompt or wrote nonsense were removed. In this study, the final sample contained 208 participants (55.3% female, M_{age} = 30.6). They were randomly allocated to a 2-mortality salience condition (mortality salience versus toothache salience) × 2-influencer types (using pet versus non-pet profile images) in a between-subjects design.

Procedure. In the first part, two questions were used to manipulate mortality salience (Dunn et al., 2020). Participants in the mortality salience condition were required to think about what their death might be like and write their comments. They were also requested to think about what might happen to their body once they were physically dead. In the toothache condition, participants were requested to write down their comments. when they experienced dental pain, then imagined what might happen to their body while experiencing dental pain.

In the subsequent section, participants were asked to complete a filler task. Previous research has indicated that individuals respond to the concept of death through two distinct mechanisms. The immediate reaction involves a denial of death. Nevertheless, heightened mortality salience can elicit distal responses, which are actions that bolster resilience or shield consumers from the awareness of death in the long term (Goldenberg et al., 2000). Therefore, following previous research (Dunn et al., 2020), we included a filler task between mortality salience and the response behaviors. Next, participants would randomly see an advertisement for snacks posted by one social media influencer (pet or non-pet). In the using pet profile image condition, the profile photo of the influencer was a pet dog, and the ID was "Pet blogger A". In the using non-pet profile image condition, the profile photo was a shopping bag,

Table 2
Scale reliability and validity.

Study	Construct	Items	Reference	Cronbach's Alpha	Average Variance Extracted	Composite Reliability
Study1a	Perceived similarity	A His/her way of thinking is similar to mine B His/her behavior is similar to mine C He/she is similar to me	Schouten et al. (2020)	0.900	0.654	0.848
	Consumer attitude-Product attitude	A Attracted B Pleasant C Likable	Sun et al. (2022)	0.864	0.533	0.694
	Consumer attitude-Purchase intentions	A Click to purchase B Forward recommendation.	Liu and Liu (2020)	0.859	0.492	0.733
Study2	Product attitude	A Attracted B Good C Pleasant D benefit E Likable	Sun et al. (2022)	0.857	0.512	0.827
	Purchase intentions	A Click to purchase B Forward recommendation.	Liu and Liu (2020)	0.828	0.711	0.831
Study3	Product attitude	A Attracted B Good C Pleasant D benefit E Likable	Sun et al. (2022)	0.893	0.605	0.820
	Purchase intentions	A Click to purchase B Forward recommendation.	Liu and Liu (2020)	0.868	0.704	0.826

and the ID was "Good selection blogger A". Participants were then instructed to rate the measures of perceived similarity with the influencer (Schouten et al., 2020) and consumer attitude which is reflected by two variables: product attitude (Liu and Liu, 2020; Sun et al., 2022) and purchase intention (Alcántara-Pilar et al., 2024) from one (totally disagree) to seven (totally agree). The questions on the demographics were at the end.

3.1.2. Results

Manipulation check: We analyzed the top three words that most frequently appeared in comments from participants in the mortality salience and the control condition, respectively. In the mortality salience condition, the most mentioned words were"scared," "terror," and "haven't," and all expressed death anxiety, whereas, in the control condition, the most mentioned words were "uncomfortable," "toothache," and "be fidgety," and all expressed pain anxiety. The manipulation of mortality salience is therefore successful in the mortality salience condition.

Two-way ANOVA tests were carried out using the mortality salience condition and influencer type as independent variables and product attitude and purchase intention as dependent variables, respectively. No main effect was shown for mortality salience and influencer type. The interaction between mortality salience condition and influencer type was found to be significant across multiple dependent variables (see Fig. 2). For product attitude, the Levene's test result (0.06, $p = 0.806$) confirms equal variances ($p > 0.05$), and the interaction effect is also significant (F (1, 204) = 7.46, $p < 0.01$). For purchase intention, the Levene's test result (3.70, $p = 0.056$) suggests that the variances are equal ($p > 0.05$), with the ANOVA indicating a strong interaction effect (F (1, 204) = 14.17, $p < 0.001$). The detailed ANOVA results are presented in Table 3. In the mortality salience group, compared to those who saw a non-pet profile image, participants who saw an advertisement endorsed by a pet profile image had significantly lower ratings of product attitude ($M_{Pet} = 4.16$; SD = 1.46 vs. $M_{non\text{-}pet} = 4.86$, SD = 1.30, $t = -2.57$, $p < 0.05$) and purchase intention ($M_{Pet} = 3.46$, SD = 1.74 vs. $M_{non\text{-}pet} = 4.42$, SD = 1.71, $t = -2.82$, $p < 0.01$). In the control condition, participants preferred pet profile pictures over the counterpart, as indicated by the ratings of purchase intention ($M_{Pet} = 4.19$, SD = 1.63 vs. $M_{non\text{-}pet} = 3.44$, SD = 1.46, $t = 2.49$, $p < 0.05$), which is consistent with previous research on the attractiveness of influencers using pet profile images (Myers et al., 2024; Zhang et al., 2023). This finding supports **H1(a)**, demonstrating that mortality salience (versus the control condition) has a negative effect on consumer attitudes toward influencers using pet profile images (versus control images—non-pet images).

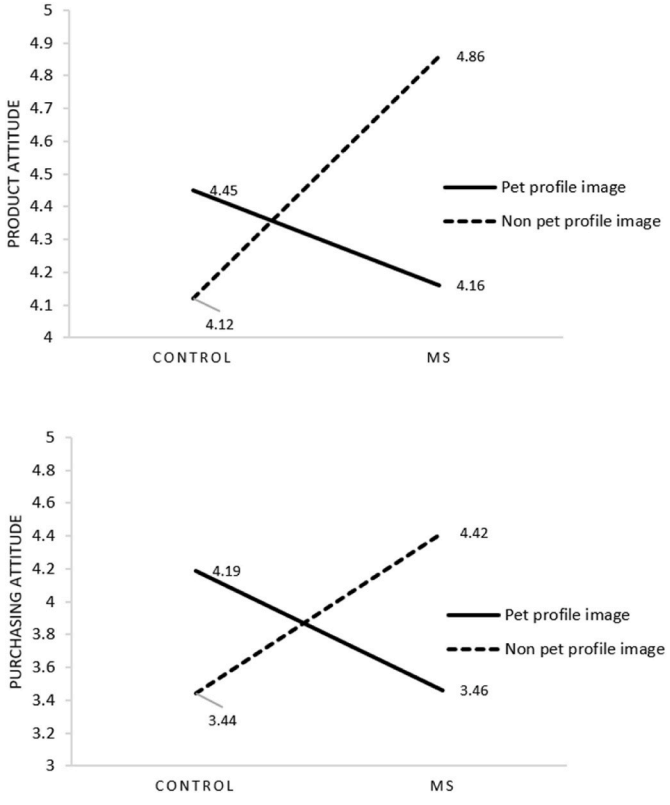


Fig. 2. Profile plot: Study 1a: Two-way ANOVA Tests of testing H1(a).

Moderated mediation analysis was run using PROCESS Model 7 (Hayes, 2018). We took mortality salience as the independent variable, product attitude as the dependent variable, perceived similarity as the mediator, and influencer type as the moderator (using 5000 bootstrap samples). As expected, the negative effect of mortality salience on product attitude was mediated by perceived similarity in the pet image condition (b = −0.4142, SE = 0.1700, 95% CI: 0.7559 to −0.0863) but not appearing in the control condition (b = 0.2370, SE = 0.1857, 95% CI: 0.1137 to 0.6084). Similarly, we used purchase intention as the dependent variable and kept the independent variable, mediating variable and moderator variable consistent with the previous ones running the PROCESS MODEL 7. The results consistently showed that the negative effect of mortality salience on purchasing intentions was mediated by perceived similarity in the pet image condition (b = −0.5911, SE = 0.2432, 95% CI: 1.0832 to −0.1256) but not appearing in the control condition (b = 0.3383, SE = 0.2629, 95% CI: 0.1697 to 0.8682; see Table 4). Therefore, the findings support **H1(b)**, demonstrating that perceived similarity mediates the relationship between mortality salience and consumer attitudes toward influencers using pet profile images.

3.2. Study 1b: IAT experiment

The IAT is a well-established experimental paradigm for detecting implicit associations, particularly unconscious or automatic attitudes that individuals may not explicitly recognize (Lecuyer et al., 2021; Wörfel, 2021). Therefore, in Study 1b, we used the IAT method to further strengthen the effect of mortality salience on consumer attitude towards pet images—H1(a). The IAT assesses these implicit attitudes and associations by analyzing response times under different conditions, offering insights into underlying biases or preferences. Specifically, following past research (Greenwald et al., 2003), consumer implicit associations were reflected by the *D*-score, which represents the difference in response times between the incongruent and congruent pairings in the following experiment.

3.2.1. Method

Participants. A total of 160 Chinese participants were recruited via another leading survey platform, Credamo. Fourteen participants were omitted from further analysis due to incomplete data, excessively rapid responses (less than 300 ms in over 10% of trials), or an excessive number of errors (accuracy rate <90%) (Greenwald et al., 2003). The final sample included 146 samples (45.9% female, Mage = 30.5).

Procedure. Participants were informed that they would first encounter a categorization task. They were randomly allocated to two groups. In the mortality salience condition, following previous research (Baka et al., 2012), participants were exposed to four pictures of death

Table 3 (a)
Study 1a -ANOVA results for product attitude.

Source	Mean square	*F*	*p*
mortality	2.598	0.187	0.740
profile_image	1.701	0.123	0.786
mortality * profile_image	13.880	7.461	0.007

Table 3 (b)
Study 1a -ANOVA results for purchase intention.

Source	Mean square	*F*	*p*
mortality	0.821	0.022	0.907
profile_image	0.529	0.014	0.925
mortality * profile_image	38.023	14.174	<0.001

Table 4
Study 1a-moderated mediation analysis results.

Mortality salience- > perceived similarity - > DV	DV: Product attitude			
	b	SE	LLCI (95%)	ULCI (95%)
Pet images	−0.4142	0.17	**−0.7559**	**−0.0863**
Non-pet images	0.237	0.1857	−0.1137	0.6084
Mortality salience- > perceived similarity - > DV	DV: Purchase intention			
	b	SE	LLCI (95%)	ULCI (95%)
Pet images	−0.5911	0.2432	**−1.0832**	**−0.1256**
Non-pet images	0.3383	0.2629	−0.1697	0.8682

and four death-related words: "plague, corpse, murder, and death". In the control condition, the bad words were "pain, tiredness, weariness, and fear," and the bad picture illustrated people in pain, tiredness, or fear. Then participants were requested to finish a filler task to elicit distal terror management effects.

Later, in both conditions, they were asked to participate in four categorized tasks: categorizing pet pictures, eight words (four good & four bad), and eight pictures (four good & four bad) into the "good" or "bad" concept as fast as possible. In block 1, pet pictures and good words were combined ("pet or good words" for the "F" key vs. "bad words" for the "J" key), and 15 practice trials were conducted by making the word 'false" appear on their screen if they were wrong. Block 2 contained 24 trials each, which were the same as in block 1, except that there was no warning if they pressed the wrong key. In block 3, pet pictures and bad words were paired ("good words" for the "F" key vs. "pet or bad words" for the "J" key), and 15 practice trials (with a warning if they were wrong) were conducted. Block 4 contained 24 trials similar to those in block 3, except for the warning part. Within each block, pictures and words were presented randomly. The *D*-score was used as the dependent variable, which represents the difference in response times between the incongruent (pet and bad are represented by the same key—Block 3 or 4) and congruent (pet and good are represented by the same key—Block 1 or 2) pairings. Higher *D*-scores indicate stronger favorable implicit associations with pet pictures (Greenwald et al., 2003).

3.2.2. Results

Results from independent-samples t-tests showed that the *D*-score is lower in the mortality salience than that in the control condition, and the difference is marginally significant ($M_{mortality\ salience} = -0.09$, $M_{Control} = 0.05$, $t = 1.86$, $p < 0.07$; See Fig. 3). This result showed participants in the mortality salience condition performed less favorable implicit associations with pet pictures than those in the control condition. Therefore, the result supported H1(a).

3.3. Study 2 The moderating role of creatureliness

3.3.1. Method

Participants. 300 participants were recruited from Wenjuanxing. Because of failure to adequately complete some tasks, 19 participants were removed from analyses, leaving a final sample of 281 (52.0% female, $M_{Age} = 29.7$). They were randomly allocated to a 2-mortality salience condition (mortality salience versus control) × 2 creatureliness (high versus low) between-subjects design.

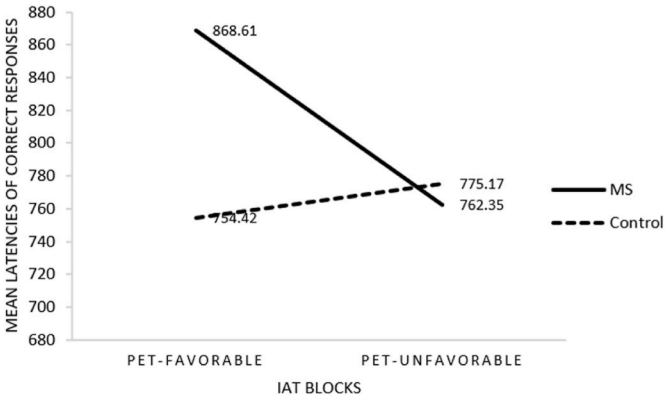


Fig. 3. Profile plot: Study1b: IAT Experiment. (Response time of latencies).

Procedure. In the mortality salience condition, referencing previous research (Lifshin et al., 2023), participants saw a black T-shirt with a skull image and the word "death." A similar blue T-shirt without a skull image was shown in the control condition. In the next stage, the creatureliness manipulation and manipulation check were introduced (Goldenberg et al., 2001). In high creatureliness condition, it emphasized that human bodies share much the same features with the other animal bodies. The essay in low creatureliness condition emphasized the uniqueness of humans. The details are shown as APPENDIX. Participants rated the animal-human continuity scale (AHCS) for manipulation check (Templer et al., 2006).The higher AHCS score indicated that participants perceived greater differences between humans and other animals. Later participants were requested to finish a filler task to elicit distal terror management effects. Finally, participants would see an advertisement for toilet paper posted by an influencer using a pet profile image. They were instructed to rate the measures of product attitude and purchase intention from one (totally disagree) to seven (totally agree). The questions of the demographics were listed in the end.

3.3.2. Results

Manipulation check: In the mortality salience condition, the most mentioned words were "terror," "scared," and "dislike," whereas, in the control condition, the most mentioned words were "image," "color," and "cloth." The manipulation of mortality salience was therefore successful. In addition, the independent sample *t*-test showed that the rating of AHCS was significantly lower in the high creatureliness condition ($M = 3.77$) than in the low creatureliness condition ($M = 3.94$, $t = 1.86$, $p = 0.06$). The manipulation of creatureliness was also therefore successful.

Tests of moderation (PROCESS Model 1) were conducted to assess H2 (Preacher et al., 2007). Creatureliness (1 = high, 0 = low) was treated as a moderator of the relationship between mortality salience condition (1 = mortality salience, 0 = control) and consumer attitude towards advertisement.

As shown in Table 5, after controlling for gender, age, education, and income, a significant interaction between mortality salience condition and creatureliness was found for attitude ($\beta = -0.70$, $p < 0.05$) and purchase intention ($\beta = -0.86$, $p < 0.05$). When a high creatureliness prime was introduced, participants reported lower attitude ($M_A = 4.30$) and purchase intention ($M_{PI} = 3.43$) in the mortality salience condition than in the control condition ($M_A = 4.74$, $M_{PI} = 4.17$, $p < 0.05$). In contrast, in the low creatureliness condition with a prime of human uniqueness, no significant difference was found between the mortality salience condition and the control condition (see Fig. 4). This supported H2.

Table 5
Study2: Model summary of PROCESS Marco Model 1.

	Product Attitude		Purchase Intention	
Predictor	β	p	β	p
Constant	2.67***	0.00	2.01**	0.01
MS condition (X)	0.13	0.48	−0.02	0.95
Creatureliness (W)	0.14	0.47	−0.01	0.97
X × W	−0.70**	0.01	−0.86*	0.02
Gender	−0.16	0.22	−0.12	0.51
Age	0.27**	0.00	0.41***	0.00
Education	0.30*	0.02	0.24	0.18
Income	0.14	0.11	0.11	0.33

*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

3.4. Study3 mortality salience, worldview disclosure, and endorsement outcome

3.4.1. Method

Participants. We recruited 331 participants from Wenjuanxing (59.5% female, $M_{age} = 29.7$). Participants were randomly allocated to 2 (mortality salience versus control) × 2 (worldview disclosure with versus without) between-subjects designs.

Procedure. Mortality salience was primed with two open questions, "Describe the thought of your own death that arouses in you" and "Imagine what will happen to you once you are physically dead." Participants in the control condition were requested to answer parallel questions about failure. Then, participants were asked to finish a filler task to elicit distal responses toward death terror. Later, participants were randomly allocated to assess an advertisement posted by an influencer using a pet profile image either with worldview disclosure (income from this ad will be partly donated to animal welfare) or without it. After that, they were instructed to finish the measures of perceived similarity with the influencer, product attitude, and purchase intention. Demographics were collected in the end.

3.4.2. Results

The study conducted a 2 × 2 ANCOVA with gender, age, education, and income as control variables. A significant main effect of worldview disclosure was found regarding product attitude ($M_{With} = 5.19$, $M_{Without}$ 4.69, F (1,330) = 6.70, $p < 0.01$). More importantly, significant interactions were identified (See Fig. 5). It was found that the interaction between mortality salience conditions and worldview disclosure conditions was significant for both product attitude (F (1,330) = 6.19, $p < 0.05$) and purchase intention (F (1,330) = 5.98, $p < 0.05$). It revealed that when there was no worldview disclosure with the advertisement, the endorsement outcomes of influencers using pet profile images would be lower in the mortality salience condition than in the control condition regarding product attitude ($M_{Mortality\ salience} = 4.51$, $M_{Control} = 4.87$, F [1, 166] = 3.59, $p < 0.06$) and purchase intention ($M_{Mortality\ salience} = 3.70$, $M_{Control} = 4.30$, F [1, 166] = 3.59, $p < 0.06$). However, when worldview disclosure was included in the advertisement, the endorsement outcomes of influencers using pet profile images would be higher in the MS condition than in the control condition regarding product attitude ($M_{Mortality\ salience} = 5.19$, $M_{Control} = 4.89$, F [1, 163] = 4.53, $p < 0.05$) and purchase intention ($M_{Mortality\ salience} = 4.60$, $M_{Control} = 4.22$, F

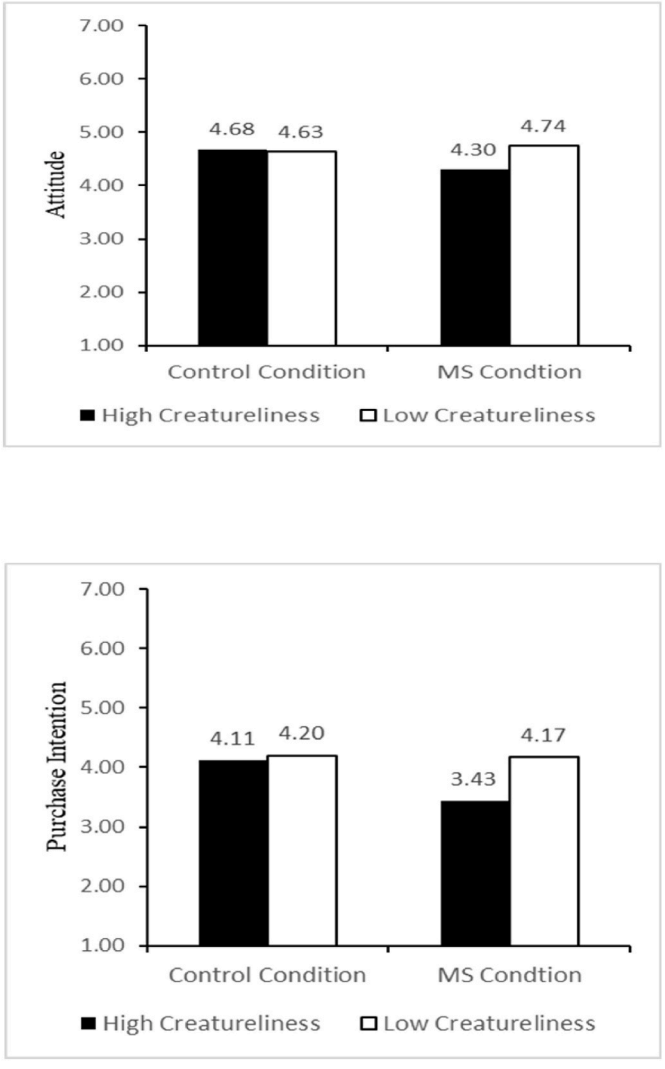


Fig. 4. Study2: Consumer attitudes toward influencers using pet profile images (High Vs. Low creatureliness).

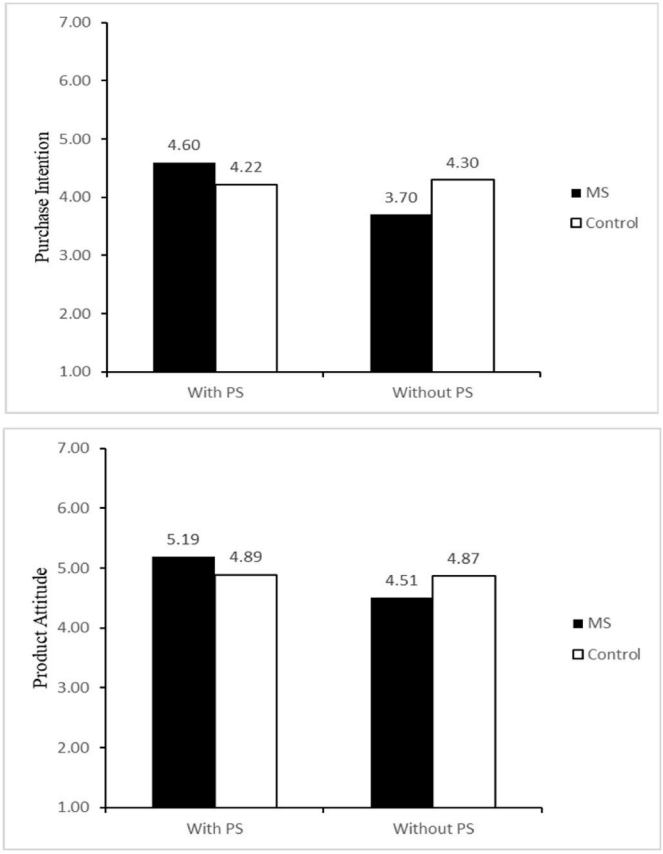


Fig. 5. Study3: Mortality salience (MS) and worldview disclosure (Example: Philanthropy statement-PS) interaction on consumer attitude.

[1, 163] = 3.92, $p < 0.05$). These results, therefore, supported H3.

4. General discussion

The current work investigated how mortality salience impacted consumer attitudes toward influencers using pet profile images and how worldview disclosure reversed this negative effect.

First, H1 was supported, based on Study 1a and Study 1b. Specifically, mortality salience negatively affected perceived similarity, product attitude and purchase intention toward products endorsed by influencers using pet profile images, while the effect of mortality salience was not significant on counterparts (Study 1a). Then we used IAT to provide additional evidence by showing participants responding significantly faster toward pet-bad pairings than pet-good pairings in mortality salience conditions (Study 1b). Previous research investigated that reminding people of mortality salience increased their need to distance from biological identity (Lifshin et al., 2023), which is consistent with our findings that mortality salience negatively affects the influencers using pet image profiles that express and convey biological information.

Second, we verified H2, which implied that the mechanism of the above effect was through perceived creatureliness—how consumers perceive similarity of biological conditions with other animals (Study 2). That is, compared with low perceived creatureliness, only in consumers with high perceived creatureliness, who emphasized similarities between humans and animals, can mortality salience negatively affect influencers using pet profile images. Previous psychological research demonstrated that humans and animals similarly had creaturely limitations, and perceiving this similarity contributes to developing emotional rejection toward vulnerability to death (Goldenberg et al., 2000; Lifshin et al., 2023). Following this, our research applied perceived creatureliness into the marketing area, being consistent with previous research.

Third, H3 was supported by Study3 that showed that worldview disclosure reverses the above negative effect as it provided consumers with worldviews (Study 3). Previous research demonstrated that worldview beliefs provide people with more valuable meaning, thereby buffering the heightened accessibility of mortal thoughts in various contexts (Lyu et al., 2023; Rahimah et al., 2020). This research is in line with previous research, which found the buffering function of worldview towards mortality salience in influencer marketing.

4.1. Managerial implications

Results show that mortality salience negatively affects influencers using pet profile images. Therefore, in times of heightened existential anxiety (e.g., during regional wars and conflicts or pandemics), we encourage influencers to pay attention to the careful selection of pet images as profile images under consideration of mortality salience. Moreover, as for the pet influencers who benefit by posting about daily events of their pets and selling pet-related products, we provide novel insights. That is, we proposed worldview disclosure, such as donation activities and animal care sponsors, contributing to reversing the negative impact of mortality salience on influencers using pet profile images based on TMT (Greenberg et al., 1992). Additionally, we encourage influencers to imply the uniqueness of individuals to consumers in their posts, which protects consumers from the negative impact of mortality salience because of perceived less creatureliness.

For brands leveraging pet influencers, integrating messages that convey existential meaning and cultural values can enhance consumer engagement, particularly in social contexts where mortality cues are prevalent. For example, campaigns could portray pets as symbols of enduring love, loyalty, or the joy of life—resonating with deeper human values that help alleviate mortality-related anxieties. Additionally, brands can adopt dynamic marketing strategies that respond to consumers' psychological states. During times of societal stress or following mortality-related events (such as natural disasters or pandemics), emphasizing themes of connection, purpose, and legacy in pet-related campaigns can foster positive emotional associations and strengthen brand loyalty.

Social media platforms should carefully manage the contextual placement of pet influencer content. Specifically, it is advisable to avoid recommending or displaying pet influencer-endorsed content alongside mortality-related news or distressing information. Our findings suggest that mortality salience cues can undermine the effectiveness of pet influencers by triggering existential anxiety, leading to psychological distancing from pet-related images. Instead, platforms could strategically position pet influencer content in contexts that foster positivity or convey meaningful worldviews, thereby enhancing consumer engagement and mitigating potential negative associations.

4.2. Theoretical contributions

The current study offers several theoretical contributions. First, our research enriches the field of influencer marketing by introducing mortality salience as a new antecedent influencing consumer behavior. While recent studies have shown that pet images effectively promote consumer engagement due to the symbolic relationships and perceived ownership people feel toward pets (D'Souza et al., 2023), these studies often overlook the impact of mortality cues—a prevalent factor in today's social media environment. Leveraging Terror Management Theory (TMT), our research indicates that being reminded of mortality can diminish the impact of pet images. Previous TMT studies have posited that awareness of mortality heightens individuals' desire to separate from their biological nature as a psychological safeguard against the anxiety associated with death (Lifshin et al., 2023; Liu et al., 2022). Our findings support this by showing that mortality salience leads consumers to feel less connected to pet images, thereby impairing their attitudes toward influencers using these images.

Second, this study advances the understanding of defensive mechanisms against mortality salience. We illustrate that when pet influencers share content aligned with consumers' worldviews—such as messages about philanthropy—It can alleviate the adverse impacts of mortality salience. This implies that consumers employ content affirming their worldview as a means to counteract existential anxiety, consistent with the broader principles of TMT, which propose that cultural worldviews serve as defenses against death anxiety (Dunn et al., 2020; Greenberg et al., 1992; Lyu et al., 2023; Zaleskiewicz et al., 2015).

Third, our work extends the application of TMT into the domain of influencer marketing—an area where this theoretical framework has been underexplored. We examine consumers' negative reactions toward influencers using pet profile images in the context of mortality salience and identify strategies to reverse these negative effects. Specifically, we find that sharing worldview-affirming content can transform consumer attitudes from negative to positive. This highlights the relevance of TMT in understanding attitudes toward animals in marketing contexts, demonstrating that existential anxiety can lead to psychological distancing from pet influencers due to perceived dissimilarity.

4.3. Limitations and future research

First, our research ignores the social resources of influencers using pet profile images when exploring the impact of mortality salience on consumer attitude towards them, such as the number of followers and historic activities (Wies et al., 2023). These can be regarded as social identity and might generate a defense mechanism towards mortality salience, thereby mitigating the original negative effect. Hence, it is meaningful for future research to explore the effect of such antecedents, providing more comprehensive practical guidance for pet influencer marketing.

Second, our research mainly focused on the short-term effect of mortality salience on consumer attitudes toward products endorsed by

influencers using pets profile images. Future research may consider the perspective of long-term effects, since it is crucial to consider the long-term relationship with consumers (Liu et al., 2017). In this regard, the key question in our context is whether and how long it would take for the negative effect to fade away, which may give deeper guidance to market practice. Additionally, if conditions permit, we also strongly encourage future research examining actual consumption of products endorsed by influencers using pet profiles in the real-world context, which will supplement the external validity of the current findings.

CRediT authorship contribution statement

Yongdan Liu: Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Matthew Tingchi Liu:** Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Yutong Ma:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104239.

Appendix. Stimuli for studies

Study1a:

Mortality salience vs. Control-pain salience conditions

Mortality salience:.

1) Please describe your thoughts when you are imagining that you are going to die.

2) Please imagine and describe in as much detail as possible what changes will occur to your body at the moment of death or after death?

Control condition:.

1) Please describe your thoughts when you are imagining that you have the toothache.

2) Please imagine and describe in as much detail as possible what changes will occur to your body at the moment of having the toothache?

Pet vs. Non-pet profile image conditions





Pet blogger A:. Aug.30 14:45 from Weibo weibo.com.
Taotaoju Honeycomb Original/Sesame Egg Roll 160g.
Click on the web link; more products are waiting for you to place an order ~
Good selection blogger A.
Aug.30 14:45 from Weibo weibo.com.
Taotaoju Honeycomb Original/Sesame Egg Roll 160g.
Click on the web link; more products are waiting for you to place an order ~

Study1b:

Mortality salience vs. Control-pain salience conditions





Study2:

Mortality salience vs. Control conditions





High vs. Low perceived creatureliness. **High**: The boundary between humans and animals is not as large as most people think. Human achievements seem to be the result of complex thinking and free will, but actually are the result of our biological instinct and simple learning experience.

Low: Although we humans have some commonalities with other animals, humans are indeed unique. We are not simply selfish creatures driven by hunger and desire, but complex individuals with our own will, able to make choices and create their own destiny.

Study3:

Mortality salience vs. Control-failure salience conditions

Mortality salience:.

1) Please describe your thoughts when you are imagining that you are going to die.

2) Please imagine and describe in as much detail as possible what changes will occur to your body at the moment of death or after death?

Control-failure salience conditions:.

1) Please describe your thoughts and emotions when you are imagining that you are going to fail.

2) Please imagine and describe in as much detail as possible what changes will occur to your body at the moment of having the failure?

With worldview disclosure vs. No worldview disclosure conditions.

With worldview disclosure:. There are some pet bloggers on social media who rely on sharing their pet's updates to gain fans, attract audiences, and bring themselves many business opportunities. Pet blogger A said that part of the income from social media advertising will be used to support Animal welfare.

No worldview disclosure conditions:. There are some pet bloggers on social media who rely on sharing their pet's updates to gain fans, attract audiences, and bring themselves many business opportunities.

Data availability

Data will be made available on request.

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